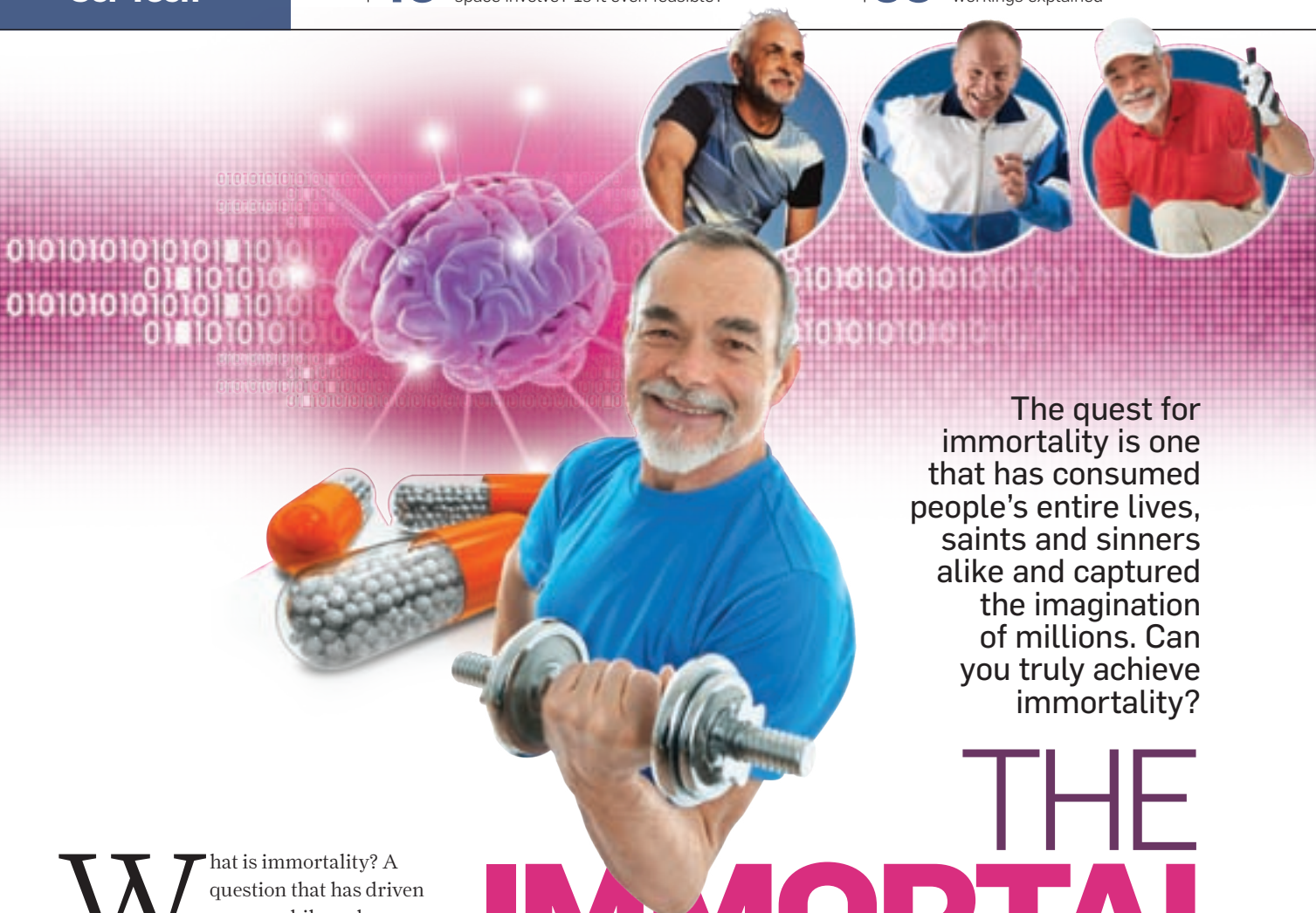




The quest for immortality is one that has consumed people's entire lives, saints and sinners alike and captured the imagination of millions. Can you truly achieve immortality?

THE IMMORTAL BEING

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What is immortality? A question that has driven many a philosopher to madness and one to which no one truly has an answer as yet. When you think of yourself as immortal, what do you envision? Your body frozen in time, but your brain untouched, gaining wisdom with the progress of time but still living and experiencing the human condition? A being of pure energy, one that can observe the passage of millenia with leisure (Dr. Manhattan, ring a bell?). Whatever your concept of immortality, we take a look at what medicine can do for us in this regard and see how far it can get us.

What is ageing?

Ageing is nothing more than your body wearing out and being unable to repair itself. An accumulation of changes. Consequently, a person who looks young at even, say, 50, does so because that person hasn't actually aged much. His or her cells are still going strong and are in better shape than those of the average

person. Scientists have a term for this, they call it negligible senescence, which is nothing more than a fancy term for saying 'negligible ageing'. Some organisms such as sturgeons and tortoises are considered to be in such a state. They are not immortal of course, but they do seem to live for hundreds of years with a barely noticeable difference in their physical condition.

Our body wears out over time with use. It is constantly bombarded with radiation, UV light, even foods such as glucose can damage the body from the inside, not to mention the action of various microbes. The cells in our body die and regenerate continuously because of this and several other factors. In fact, every few seconds,

approximately 50 million cells die and are regenerated and thus, a healthy body is one in which the cells are regenerating continuously. Once your rate of regeneration slows down and is unable to keep up with the rate of cell death, that is when you begin to show signs of ageing. Most cells in your body are only a few hours old, many even younger. Humans do experience a peak of regeneration or a state of "low senescence" between the ages of 20-35. A condition, if maintained, can lead to an exceptionally long life.

The basics

To understand ageing, we need to understand how a cell grows old. Cells in our